

Session 8 — Indian sky, and what to do next

Practical Astronomy Intensive

7 questions 15 minutes 5 concepts 7 diagnostic

Positional astronomy, not folklore: $360/27 = 13^\circ 20'$, 12° of elongation per tithi, five limbs from two longitudes. Plus the recognition that Saptarishi is the Dipper they already found — and that the camp has an afterwards.

- 1 On night one you found the Big Dipper and Polaris. Tonight someone mentions Saptarishi and Dhruva. What are they pointing at?

indian-star-names

Recall

Multiple choice

- a A different pole star from Polaris, used in the Indian system
REVEALS Dhruva believed to be some pole star other than Polaris
- b** The same two things — Saptarishi IS those seven stars, and Dhruva IS Polaris; only the names and the stories differ ✓
- c Two different patterns in another part of the sky, which you have not found yet
REVEALS Saptarishi and the Big Dipper believed to be different star patterns
- d The Sanskrit translations of the Greek names, coined after Greek astronomy reached India
REVEALS Indian star names assumed to be translations of Greek ones — the naming is independent and often older

One sky, two map legends. The naming is independent, not derivative, and sometimes finer: Arundhati–Vasishtha marks the Mizar–Alcor pair as a couple — a naked-eye double you can still check tonight — which Greek astronomy never named as a pair at all.

- 2 The ecliptic gets divided into 12 rashis and also into 27 nakshatras. What is a nakshatra, precisely?

nakshatra

Apply

Multiple choice

- a One of 27 divisions chosen because 27 is a traditionally auspicious number
REVEALS 27 assumed arbitrary rather than set by the Moon's 27.3-day sidereal circuit
- b** A fixed $13^\circ 20'$ segment of the ecliptic — $360/27$ — cut to the Moon's daily step of about 13° against the stars, and named for a marker star (yogatara) that lies in it ✓
- c A constellation, whose boundaries follow the shape of the star pattern it is named for
REVEALS Nakshatra taken as a constellation rather than a fixed $13^\circ 20'$ ecliptic segment
- d The 12 rashis renamed in Sanskrit — the same divisions under different labels
REVEALS 27 nakshatras believed to be the 12 rashis renamed — they are a different division entirely

The arithmetic is the argument. The Moon returns to the same star in about 27.3 days, so it advances roughly 13° per night — divide 360° by 27 and each segment is one night's travel. The rashis divide the same ecliptic by 12, cut to the Sun's monthly step. Different divisor, different body, same circle.

3 A tithi is often glossed in English as 'lunar day'. How long is one, and what sets its length?

tithi Apply Multiple choice

- a Exactly 24 hours — tithi is simply the Indian word for a day
REVEALS Tithi taken as a synonym for a civil day rather than an angle-defined interval
- b Exactly one thirtieth of 30 days, since 30 tithis make a month
REVEALS 30 tithis assumed to make a 30-day month — they make one synodic month, about 29.5 days
- c Exactly 12 hours, one for each degree of elongation gained
REVEALS Tithi length confused with the 12° definition itself
- d Anywhere from about 19 to 26 hours — a tithi ends when the Moon has gained 12° of elongation on the Sun, and the Moon's speed varies around its elliptical orbit ✓

This is the key structural point of the Indian calendar: a tithi is an ANGLE interval, not a time interval. $30 \times 12^\circ = 360^\circ$, one new moon to the next, about 29.5 days. Because the Moon's angular speed varies, the tithis fill that unevenly — hence the 19-to-26-hour spread.

4 A panchanga lists five quantities each day: vara, tithi, nakshatra, yoga, karana. What is the document?

panchanga Apply Multiple choice

- a A computed almanac — all five limbs are functions of the Sun's and the Moon's longitudes on that day, which is to say an ephemeris in cultural form ✓
- b An astrological text, since that is what people consult it for
REVEALS Panchanga taken as astrology rather than a computed almanac; the astrological reading is one use of the table, not the table
- c A table copied forward from classical texts, preserved unchanged for accuracy
REVEALS Panchanga assumed copied forward — a drik panchanga is recomputed from actual positions
- d Five separate traditions bundled into one book over the centuries
REVEALS Five limbs assumed unrelated — all five come from the same two longitudes

Take the Sun's longitude and the Moon's longitude: tithi is their difference in 12° steps, nakshatra is the Moon's longitude in 13°20' steps, yoga is their sum in 13°20' steps, karana is half a tithi, and vara is the weekday. Five outputs, two inputs. A drik ('observed') panchanga recomputes these from actual positions each year and openly disagrees with the older approximations — a data table that publishes its own corrections.

5 The camp ends tomorrow. A student owns binoculars, no telescope, and lives under a suburban sky. What can she realistically contribute to actual research?

citizen-science Recall Multiple choice

- a A great deal — AAVSO variable-star estimates from binoculars, IMO meteor counts by naked eye, occultation timings, or Zooniverse classifications all feed straight into professional work ✓
- b She can observe for her own enjoyment, but amateur numbers are a hobbyist exercise nobody really uses
REVEALS Amateur measurements dismissed as unused — AAVSO light curves and occultation timings feed real papers
- c Very little on her own — one observer's handful of estimates cannot matter against professional data
REVEALS One observer's contribution thought negligible — these programmes work by pooling many modest observations
- d Nothing yet — she needs a telescope and a CCD before her measurements would be worth submitting
REVEALS Expensive kit believed necessary to contribute — a naked-eye meteor count is real data

These programmes exist precisely because they buy what money cannot: coverage. No professional telescope gets time to watch one variable star every clear night for thirty years, and hundreds of amateurs do exactly that. The pooling is the method, not a consolation prize.

6 Arundhati–Vasishtha is the classic naked-eye test in the Indian sky. Which pair of stars is it, and where do you look?

indian-star-names

Apply

Multiple choice

a Two stars of the Dipper's bowl, named as a pair only after telescopes revealed them

REVEALS Indian star names assumed to postdate telescopic astronomy

b Dhruva and its faint companion, near the celestial pole

REVEALS Arundhati–Vasishtha misplaced onto Polaris; the pair is in the Dipper's handle

c A pair in Orion, corresponding to the Greek naming of the belt

REVEALS Indian star names assumed to be translations of Greek ones, mapped onto Greek asterisms

d Mizar and Alcor — the middle star of the Big Dipper's handle, which resolves into two for anyone with good eyes and a dark sky ✓

Mizar (ζ UMa) and Alcor, about 12 arcminutes apart, in the Dipper's handle. It doubles as an eyesight and transparency test, and it is a genuinely independent observation: the pair was named as a couple in the Indian tradition long before anyone put a lens on it.

7 A panchanga shows a tithi that begins after sunrise and ends before the next sunrise, so no sunrise falls inside it — that date's tithi is skipped (kshaya). How is that possible?

tithi

Reason

Multiple choice

a The Moon was invisible that day, so the tithi could not be observed and was dropped

REVEALS Tithi treated as an observation rather than a computed angular interval

b Tithis are all 24 hours, so the skip must come from a leap-day adjustment in the civil calendar

REVEALS Tithi taken as a fixed 24-hour day, with the mismatch blamed on the civil calendar

c It is a computational error in that panchanga — a date always carries exactly one tithi

REVEALS Each date assumed to carry exactly one tithi — kshaya and adhika tithis are real

d Tithis vary in length, so a short one (near 19 hours) can start and finish entirely between two sunrises, while a long one (near 26 hours) can span two sunrises and repeat (adhika) ✓

Kshaya and adhika tithis fall directly out of the definition and are computed, not fudged. If tithis were fixed-length they could not do this. This is what an angle-based calendar costs — and it is also the check that a student has actually understood the definition rather than memorised the word.